

Scientific Programming with Python

Assignment: NumPy Review

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Goal The goal of this assignment is to review the NumPy library [1]. You will interact with a Generative AI tool (acting as a personal tutor) to solidify your understanding of NumPy and Python [2, 3]. In keeping with the core learning objectives of this course, you are expected to develop the code solutions on your own, ensure your code is concise, document your thought process, and properly cite existing knowledge.

Assignment Tasks

Task 1 Use a Generative AI tool (e.g., via KI:connect) to help you review NumPy. Instruct the AI to act as a tutor and request that it generate 1 or 2 small coding exercises for you to solve. For example, the topics of your conversation could involve the following:

- (a) Basic NumPy functions and array operations (e.g., shape manipulation, indexing, slicing, filtering),
- (b) Creating and concatenating arrays of different sizes,
- (c) Performing a speed/performance comparison between a standard Python loop and a vectorized NumPy operation,
- (d) Defining and manipulating polynomial functions using the `numpy.polynomial.polynomial` module,
- (e) Generating noisy toy data and applying polynomial curve fitting via `Polynomial.fit`.

Deliverables for your notebook: Include the exact prompt(s) you gave the AI, the coding task(s) the AI provided you, and your independent Python code solving those tasks.

Allowed Python3 [2, 3] functions & libraries/modules

- All built-in functions and modules/libraries
- NumPy [1] library
- Pandas [4, 5] library (if desired)
- timeit library (if desired)
- typing library

Assignment Due Turn in your solution as a Jupyter-notebook to **LEA** by **June 22nd, 2026 at 13:30**.

Note 1 : Please include your **SciPro ID** at the **top** of your notebook.

Note 2 : You **do not** need to consider **significant figures**.

References

- [1] Harris, C.R., Millman, K.J., van der Walt, S.J. et al. Array programming with NumPy. *Nature*, 585 (2020) 357–362 (DOI: 10.1038/s41586-020-2649-2)
- [2] Python Software Foundation. Python Language Reference, version 3.14. Available at <http://www.python.org>. Online; accessed on June 12, 2026.
- [3] van Rossum, G. Python tutorial, Technical Report CS-R9526, Centrum voor Wiskunde en Informatica (CWI), Amsterdam, 1995.
- [4] The Pandas Development Team pandas-dev/pandas: Pandas Zenodo, 2020 (<https://pandas.pydata.org>)
- [5] Pandas user guide, https://pandas.pydata.org/docs/user_guide/index.html#user-guide. Online; accessed on June 13, 2026.